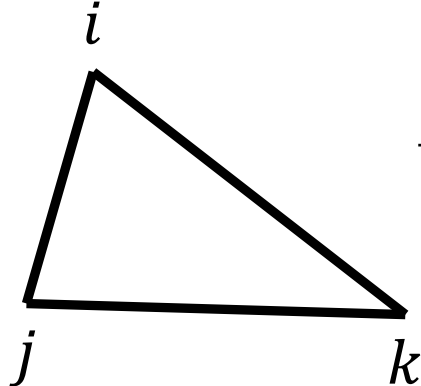
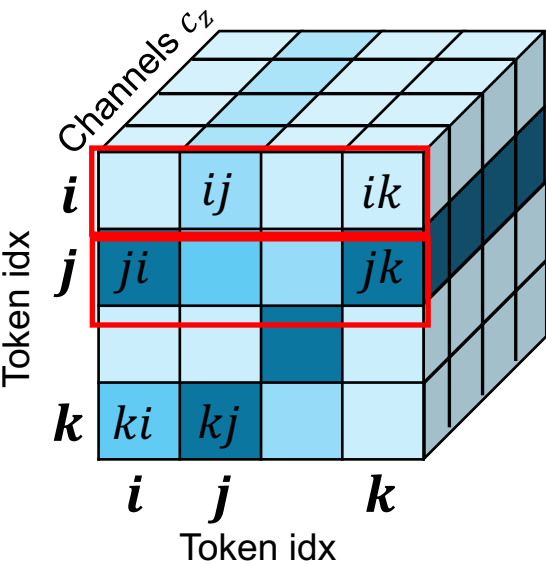




$$\vec{ij} \leq \vec{ik} + \vec{jk} \quad \text{Triangle Inequality}$$

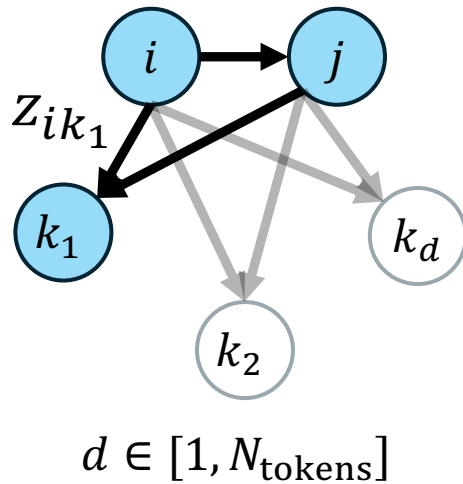
Pair representation z



Pair representations
are directional

$$z_{ij} \neq z_{ji}$$

Updating z_{ij} using
info. from outgoing edges



How do i and j tokens
relate to each k token ?

$$d_{ij} < d_{ik_1} + d_{jk_1}$$

$\vdots$

$$d_{ij} < d_{ik_d} + d_{jk_d}$$

$$d_{ij} < \sum_k (d_{ik_d} + d_{jk_d})$$

$$d_{ij} < \sum_k (d_{ik_d} \odot d_{jk_d})$$